

NORTHWEST ARKANSAS, AR, USA

WMO: 723436

Lat: 36.283N

Lon: 94.300W

Elev: 392

StdP: 96.70

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53922

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-11.6	-8.9	-17.5	0.8	-9.9	-14.7	1.1	-6.1	11.4	10.9	10.5	10.6	3.4	350	0.477

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.6	23.4	33.7	23.7	32.4	23.6	25.6	31.5	25.0	30.8	24.4	30.1	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.9	28.8	23.3	18.9	28.0	22.6	18.2	27.4	81.0	31.4	78.1	30.7	75.6	30.2	30.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
10.2	8.9	8.1	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-16.2	37.1	3.5	2.0	-18.7	38.6	-20.7	39.8	-22.6	40.9	-25.2	42.4
			-16.7	26.9	3.3	1.2	-19.1	27.7	-21.0	28.4	-22.9	29.0	-25.3	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.6	2.5	4.1	9.7	14.4	18.9	23.8	25.8	25.4	21.4	14.9	9.1	4.0
	DBStd	9.49	6.21	6.31	5.70	4.80	4.16	2.78	2.72	2.93	3.66	4.83	5.20	5.36
	HDD10.0	804	242	182	76	17	2	0	0	0	0	12	77	198
	HDD18.3	2201	490	398	271	131	45	1	0	1	13	128	278	446
	CDD10.0	2471	10	18	66	148	277	415	491	477	343	165	49	11
	CDD18.3	828	0	1	3	13	62	167	233	219	107	22	2	0
	CDH23.3	8074	0	3	21	111	497	1573	2461	2229	984	185	11	0
	CDH26.7	3353	0	0	1	14	111	608	1164	1056	363	36	1	0
Wind	WSAvg	3.5	3.9	4.1	4.3	4.3	3.6	3.1	2.6	2.6	2.6	3.2	3.8	3.7
Precipitation	PrecAvg	1179	72	62	93	114	145	127	94	76	122	93	101	75
	PrecMax	1650	187	155	262	208	260	273	287	162	280	239	228	166
	PrecMin	873	7	8	19	49	48	25	25	6	10	35	13	12
	PrecStd	191	44	41	57	50	51	62	54	39	60	48	65	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.3	22.5	25.6	28.4	31.2	35.2	37.6	38.3	33.8	29.5	24.8	20.1
		MCWB	13.9	15.7	17.1	18.6	21.3	23.4	22.8	23.3	22.6	21.3	17.3	15.3
	2%	DB	17.8	19.5	23.1	26.2	29.2	33.1	35.6	35.9	32.3	27.6	21.7	17.7
		MCWB	13.3	13.5	15.9	17.7	21.1	23.8	23.3	23.5	22.7	20.1	15.9	13.7
	5%	DB	15.6	17.2	21.1	24.3	27.8	31.6	33.8	33.8	30.7	25.4	19.9	15.2
		MCWB	10.7	12.4	14.6	17.0	21.0	23.5	24.0	23.5	22.3	18.3	14.9	11.3
	10%	DB	12.5	14.7	19.0	22.6	26.3	30.2	32.3	32.1	28.8	23.1	17.8	12.9
		MCWB	8.4	10.4	13.3	16.0	20.0	22.9	24.1	23.4	21.4	17.6	13.5	9.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	16.4	17.2	18.6	20.8	23.7	26.0	26.7	26.0	25.1	22.7	19.6	17.1
		MCDB	18.5	20.7	23.3	25.8	28.1	31.8	32.5	31.8	31.4	27.7	23.2	19.1
	2%	WB	14.7	15.3	17.2	19.5	22.7	25.0	25.8	25.3	23.9	21.6	17.5	14.7
		MCDB	17.2	18.1	21.7	23.9	27.3	30.8	31.5	31.5	29.7	26.1	20.0	16.9
	5%	WB	11.5	13.4	15.8	18.4	21.8	24.2	25.2	24.6	23.0	20.4	15.9	12.4
		MCDB	14.3	16.1	19.7	22.5	26.4	29.6	30.8	30.9	28.5	23.4	18.8	14.4
	10%	WB	8.7	10.7	14.6	17.4	20.9	23.6	24.6	24.0	22.3	19.0	14.3	10.3
		MCDB	11.7	13.9	18.2	21.3	25.2	28.7	30.1	30.1	27.2	21.9	17.1	12.4
Mean Daily Temperature Range	5% DB	MDBR	11.2	11.3	12.0	12.2	11.2	11.0	11.5	11.6	12.1	12.0	11.9	10.7
		MCDBR	14.7	14.3	14.5	13.8	12.4	12.2	13.2	13.8	13.5	13.8	13.9	13.5
		MCWBR	10.1	9.7	8.5	7.7	6.1	4.9	4.2	4.2	5.3	7.0	8.9	9.7
	5% WB	MCDBR	12.2	12.1	11.7	11.5	10.8	10.8	11.0	11.6	11.5	10.8	11.9	11.6
		MCWBR	10.0	9.6	8.0	7.5	5.9	5.0	4.7	4.7	5.3	6.4	9.3	9.8
Clear-Sky Solar Irradiance	taub	0.302	0.305	0.339	0.377	0.420	0.421	0.442	0.443	0.386	0.343	0.312	0.291	
	taud	2.485	2.487	2.423	2.328	2.259	2.280	2.275	2.249	2.376	2.481	2.521	2.536	
	Ebn at Noon	898	939	929	905	866	862	841	833	867	881	880	891	
	Edh at Noon	82	92	108	125	137	134	133	134	112	92	79	74	
All-Sky Solar Radiation	RadAvg	2.46	2.99	4.02	5.07	5.63	6.43	6.38	5.79	4.94	3.67	2.64	2.09	
	RadStd	0.22	0.42	0.24	0.29	0.48	0.57	0.39	0.39	0.37	0.47	0.29	0.27	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (28 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page